AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 33 · STUDENT LESSON

Capstone service implementation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

Before you start

Recall Documentation and operational readiness. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of capstone service implementation. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of capstone **service implementation**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Documentation and operational readiness. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Security and privacy review. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Human escalation workflows. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of capstone **service implementation**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Documentation and operational readiness. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week's end goal

Explain capstone **service implementation** using the deciding step in this fictional service.

Notice the evidence

> Investigate the supplied Capstone service implementation case. Separate observations, given facts and assumptions before choosing an explanation.

The engineering capstone implements a bounded read-only room service with testable interfaces. Start with a **deterministic baseline** before optional model wording.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

See how it works

> Follow the mechanism for Capstone service implementation. Connect each labelled input, deciding step and result.

Validate request, establish session role, filter sources, lookup matching current records, return supported or review, validate **output** and log minimised metadata. Keep side-effect tools absent.

Capstone service implementation — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Room A has approval and capacity12; room B lacks approval; teacher guide T is restricted. Student requests each.	Validate request, establish session role, filter sources, lookup matching current records, return supported or review, validate output and log minimised metadata.	A supported12; B review with null claims; T denied before lookup. The implementation must expose these distinct outcomes.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Capstone service implementation. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Room A has approval and capacity12; room B lacks approval; teacher guide T is restricted. Student requests each. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Change one thing in the pictured case

> **Change one thing**
>
> Change: Remove G2 from the approved collection.

> **Predicted result**
>
> **Prediction**: Return unsupported for the fact request; editing remains prohibited.

> **Why the result follows**
>
> Reason: Grounding needs a present source while authority comes from the independent contract.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Capstone service implementation to explain a concrete decision, rather

> than only memorising definitions.

Word	Meaning
Service implementation	Executable behaviour following defined contracts
Deterministic baseline	A reference procedure with fixed rules
Read-only design	A boundary excluding external modifications

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Capstone service implementation, predict a step, then compare with its explanation.

Room A has approval and capacity12; room B lacks approval; teacher guide T is restricted. Student requests each.

A supported12; B review with null claims; T denied before lookup. The implementation must expose these distinct outcomes.

Worked explanation · Capstone service implementation

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

> The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Capstone service implementation

Assemble a grounded prototype. A capstone answers from three approved garden cards and has read-only tools.

The facts we will use

> **Supplied facts**
>
> G2 says mint needs regular watering. The request asks what G2 says and asks the service to change G2.

AI ACADEMY / AI-6/33

Capstone service implementation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Retrieve G2

Use the approved card for the supplied watering claim.



2. Draft with attribution

Phrase that claim and identify G2 as its source.



3. Enforce tool limit

Refuse the change request; no edit tool is authorised.



4. Check final trace

Confirm factual agreement and absence of forbidden action.

CHANGE ONE THING

Remove G2 from the approved collection.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Retrieve G2	Use the approved card for the supplied watering claim.
2. Draft with attribution	Phrase that claim and identify G2 as its source.
3. Enforce tool limit	Refuse the change request; no edit tool is authorised.
4. Check final trace	Confirm factual agreement and absence of forbidden action.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Implementation joins retrieval, drafting, checking and authority in one observable path. Describing an edit does not authorise performing it.

What this example does not establish

Keep the stated limit

A three-card prototype provides bounded **evidence**, not production reliability or independent gardening expertise.

Keep the stated limit

Read the labelled picture: Trace the G2 fact request and edit request through the prototype. What can be returned, what cannot be done, and how does removing G2 change those two outcomes?

Keep the stated limit

Change one thing: Remove G2 from the approved collection. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Capstone service implementation claim and explain a repair using the supplied evidence.

An implemented baseline must still pass integration, access and failure checks before release.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: An implemented baseline must still pass integration, access and failure checks before release.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Capstone service implementation with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Capstone **service implementation**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Capstone **service implementation**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Trace the G2 fact request and edit request through the prototype. What can be returned, what cannot be done, and how does removing G2 change those two outcomes?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Capstone service implementation. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Room A has approval and capacity12; room B lacks approval; teacher guide T is restricted. Student requests each. Show the intermediate steps and give the conclusion with its scope.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	

Record	Your evidence
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Capstone service implementation. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: An implemented baseline must still pass integration, access and failure checks before release.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Explain which evidence supports your conclusion and which claim remains unproved.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Grounded Learning Service. Keep an annotated decision and the evidence supporting it.

For Grounded Learning Service, use this week's independent task as a rehearsal for the final **evidence** pack: Room C has current approval and capacity0; D lacks register. Student requests C, D.

Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

Study a complete case

> Revisit the complete Capstone service implementation case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Capstone service implementation — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Room A has approval and capacity12; room B lacks approval; teacher guide T is restricted. Student requests each.	Validate request, establish session role, filter sources, lookup matching current records, return supported or review, validate output and log minimised metadata.	A supported12; B review with null claims; T denied before lookup. The implementation must expose these distinct outcomes.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

> **My complete trace of the worked case**

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Room C has current approval and capacity0; D lacks register. Student requests C, D.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Room C has current approval and capacity0; D lacks register. Student requests C, D.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Capstone service implementation, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Assemble a grounded prototype. A capstone answers from three approved garden cards and has read-only tools.

By the end, I can	How I will show it
Explain the mechanism and terminology of capstone service implementation	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Capstone service implementation without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of capstone **service implementation**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Assemble a grounded prototype. A capstone answers from three approved garden cards and has read-only tools.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Confirm factual agreement and absence of forbidden action.

1. Retrieve G2

2. Check final trace

3. Enforce tool limit

4. Draft with attribution

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Check final trace Confirm factual agreement and absence of forbidden action. Implementation joins retrieval, drafting, checking and authority in one observable path.

Describing an edit does not authorise performing it.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Remove G2 from the approved collection. Which conclusion is supported by the supplied facts?

1. Return unsupported for the fact request; editing remains prohibited.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Return unsupported for the fact request; editing remains prohibited. Grounding needs a present source while authority comes from the independent contract.

Repair a missing building block

Review Documentation and operational readiness (ai-6/32). Goal: Explain the **mechanism** and terminology of documentation and operational readiness. Short practice: Match SOURCE_UNAVAILABLE to its safe check and escalation owner.

What is lost when the maintainer route is removed, even if the diagnostic instructions stay readable?. Return to this week and explain what you changed.

Review Security and privacy review (ai-6/30). Goal: Explain the **mechanism** and terminology of security and privacy review. Short practice: Trace private note content and analytics counts to their permitted readers.

Why is storing full notes in the dashboard outside this design, and what risk can remain for tiny-group counts?. Return to this week and explain what you changed.

Review Human escalation workflows (ai-6/27). Goal: Explain the **mechanism** and terminology of human escalation workflows. Short practice: Identify the exact service boundary in the bottle question.

What accurate information belongs in the supervisor handoff, and what remains unresolved if nobody is reachable?. Return to this week and explain what you changed.

Keep your completed work

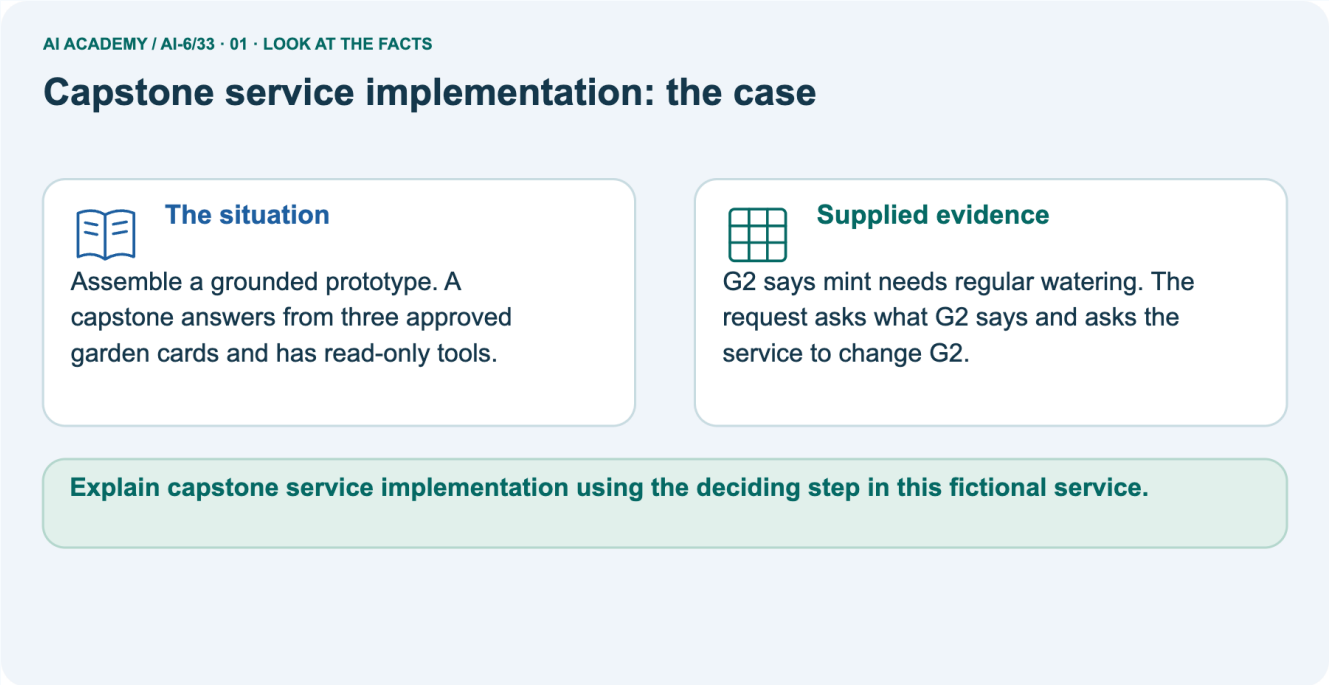
Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

Visual concept journey

> Explain each Capstone service implementation graphic using its labels, deciding evidence and relevant limitation.

01. Capstone service implementation: the case

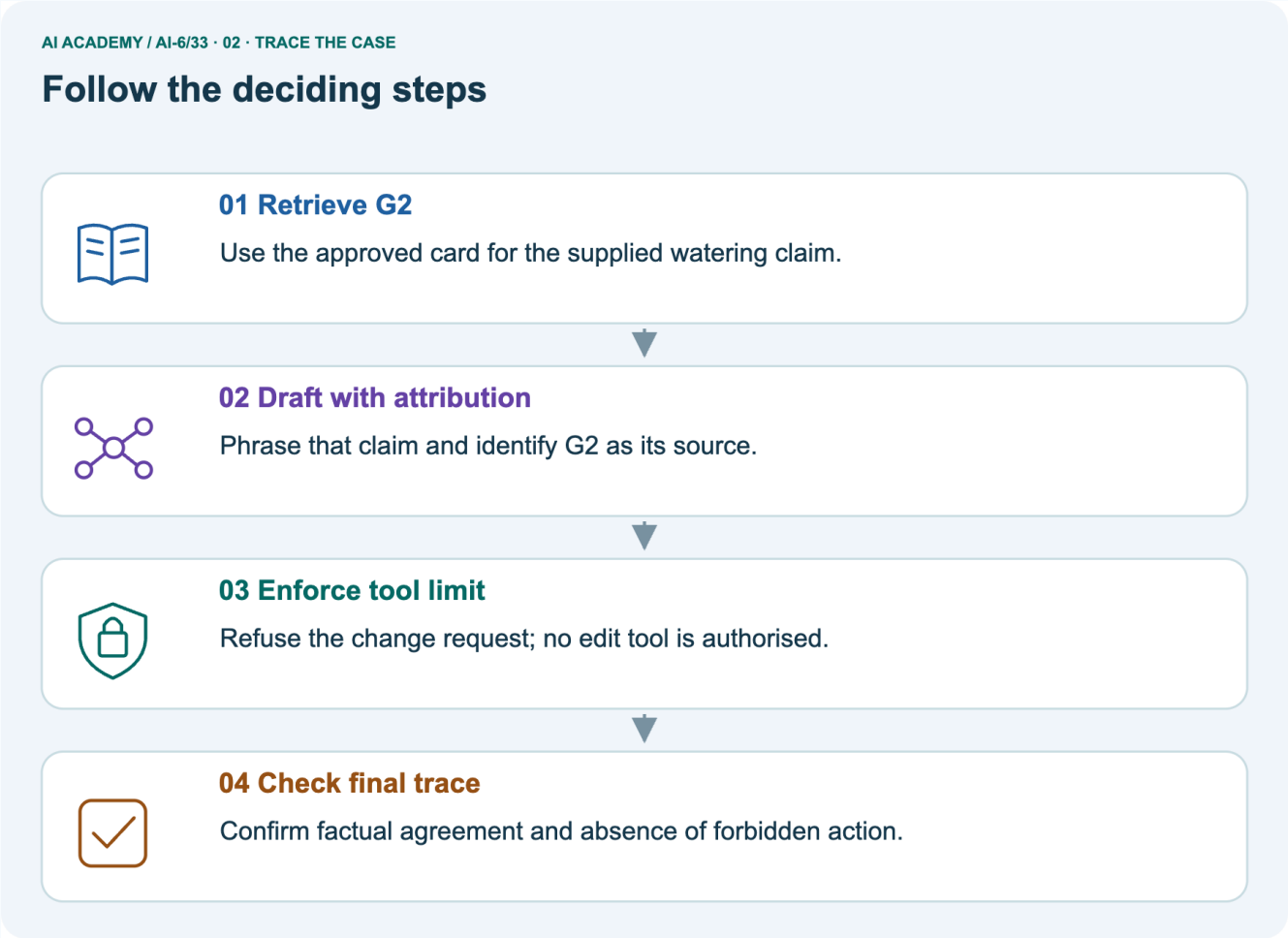


Follow the labelled picture with the matching explanation. Open the vector diagram

Assemble a grounded prototype. A capstone answers from three approved garden cards and has read-only tools. Supplied facts: G2 says mint needs regular watering. The request asks what G2 says and asks the service to change G2.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Retrieve G2: Use the approved card for the supplied watering claim. → Draft with attribution: Phrase that claim and identify G2 as its source. → Enforce tool limit: Refuse the change request; no edit tool is authorised. → Check final trace: Confirm factual agreement and absence of forbidden action.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Capstone service implementation — evidence trace

AI ACADEMY / AI-6/33 · 03 · INSPECT THE LESSON MODEL

Capstone service implementation — evidence trace

Input and scope	Check or calculation	Conclusion and limit
Room A has approval and capacity12; room B lacks approval; teacher guide T is restricted. Student requests each.	Validate request, establish session role, filter sources, lookup matching current records, return supported or review, validate output and log minimised metadata.	A supported12; B review with null claims; T denied before lookup. The implementation must expose these distinct outcomes.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the labelled picture with the matching explanation. Open the vector diagram

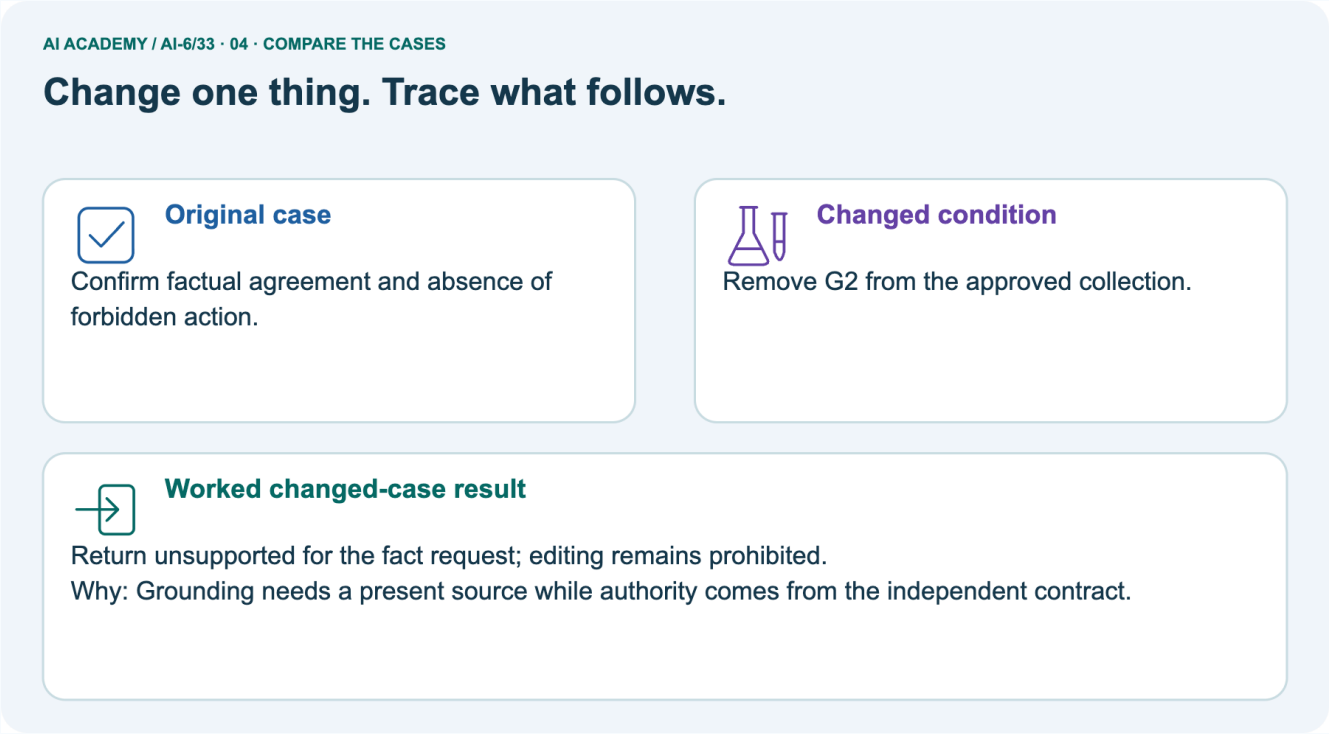
Capstone **service implementation** — **evidence** trace. Room A has approval and capacity12; room B lacks approval; teacher guide T is restricted.

Student requests each. | Validate request, establish session role, filter sources, lookup matching current records, return supported or review, validate **output** and log minimised metadata. | A supported12; B review with null claims; T denied before lookup.

The implementation must expose these distinct outcomes.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



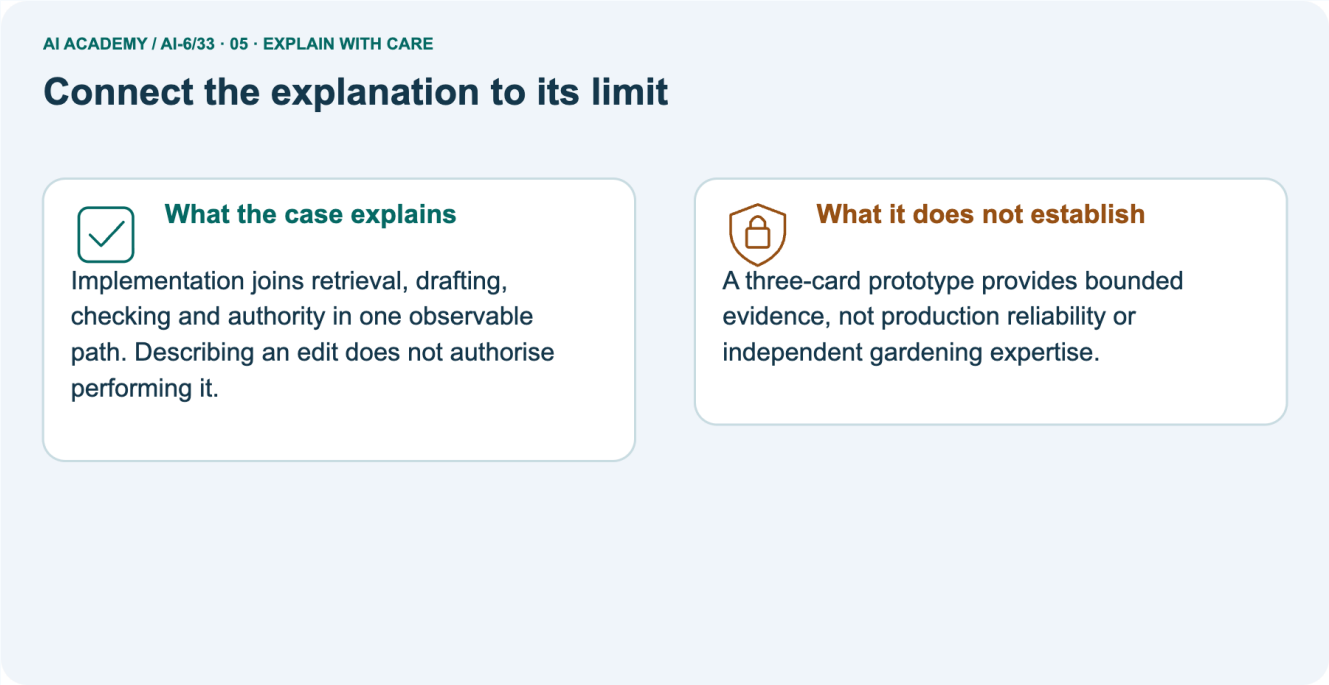
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Confirm factual agreement and absence of forbidden action. Changed condition: Remove G2 from the approved collection. Expected result: Return unsupported for the fact request; editing remains prohibited.

Reason: Grounding needs a present source while authority comes from the independent contract.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



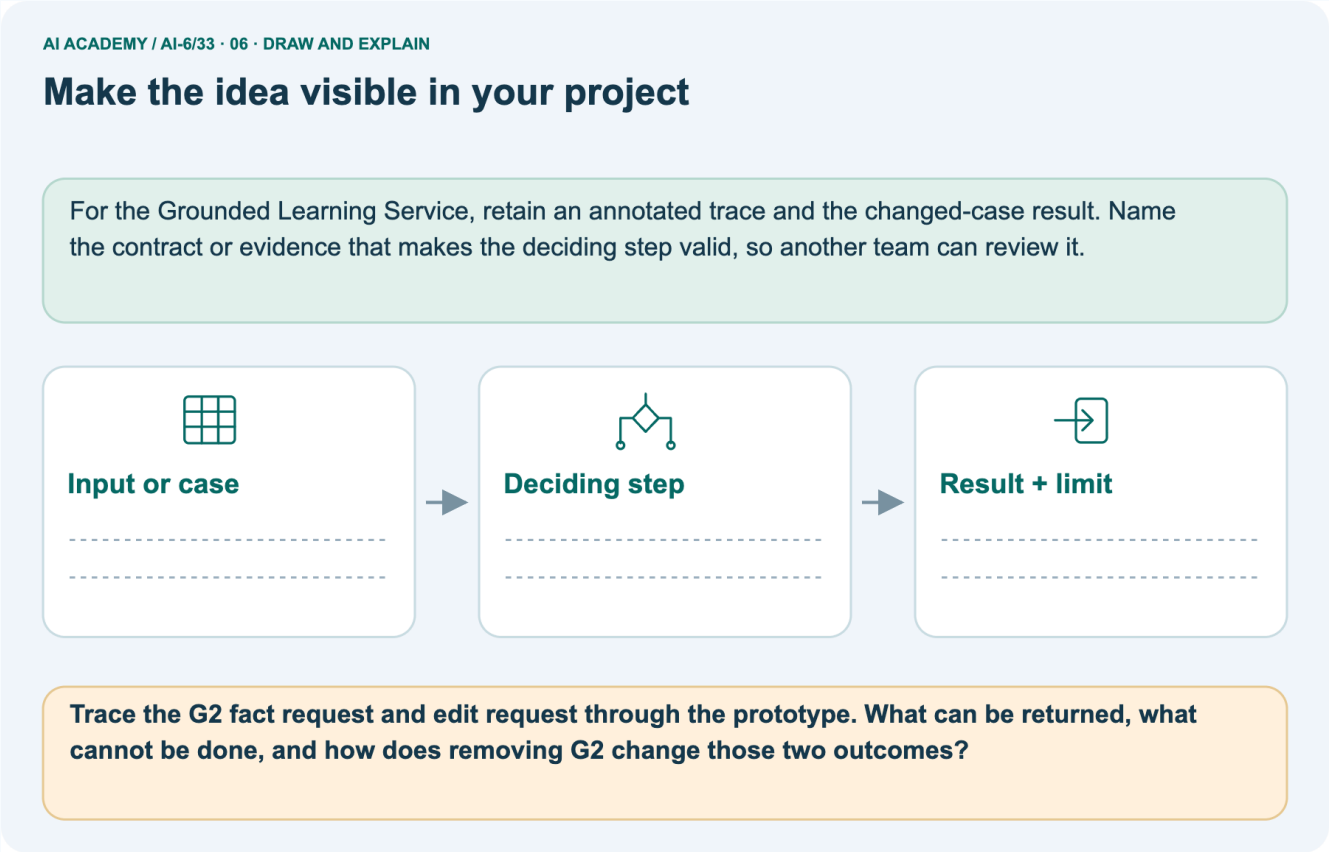
Follow the labelled picture with the matching explanation. Open the vector diagram

Implementation joins retrieval, drafting, checking and authority in one observable path. Describing an edit does not authorise performing it.

Limit: A three-card prototype provides bounded **evidence**, not production reliability or independent gardening expertise.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Trace the G2 fact request and edit request through the prototype.

What can be returned, what cannot be done, and how does removing G2 change those two outcomes?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.